Introduction to Ethereum + Lido

Crypto Leftists – Feb 5, 2025

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Today’s Agenda

No.	Step
1	PoS and nodes in Ethereum
2	DVT
3	Lido CSM
4	Q&A

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PoS and nodes in Ethereum

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Proof of Stake on Ethereum

A “security deposit” of 32 ETH is used to protect validator duties.

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Validator Rewards

Type	Source	Annual Rate
Consensus: **Issuance**	- Issued by the Ethereum network as inflation - Offset by base fee paid by users	2.4%
Consensus: **Sync Committee**		0.3%
Execution: **Transaction tips**	Direct fee paid by users for speeding up their transactions	0.3%
Execution: **MEV**	Indirect fee paid by users for “leaving money on the table”	0.3%
Total		**3.3%**

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Validator Penalties

Type	Causes	Impact	How to prevent
Missed duties	Going offline	**Low:** ~80% of what you would otherwise have earnt as rewards	- Keep your validator node synced & running 24/7 - Implement a monitoring & alerts system
Slashing	- Double signing - “Surrounding” attestations ***due to client bugs***	**High:** 1/32 of your total effective balance + leakage (<=0.2 ETH per key)	- Don’t use the same validator key in more than 1 setup - Keep your keys secure - Keep your clients updated - Follow slashing protection best practices
Correlation penalties	Being penalised along with a large proportion of the validator network	**Very High:** Up to your entire stake	- Avoid using supermajority clients or VMs with a large market share

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Process Transactions

Proposer: “Hey guys, I checked this transaction and it ***doesn’t contradict with my records***–Let’s add it to the blockchain”

Attesters: “Ok, we checked and can confirm that it ***doesn’t contradict with any previous transactions*** in our records”

Consensus *on validity of transaction* ***is reached*** *when >⅔ agree*

…

…

Secure Transactions

After reaching consensus twice, this transaction is now **finalised**

Attempts to **violate finality** will result in **slashing**

Need to ***burn ⅓ of all staked ETH*** *to attempt reversion of finalised transactions*

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Solo staker percentage today = ~6.5%

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More solo stakers =

More independent verifiers & enforcers =

Greater security, resilience, & censorship resistance

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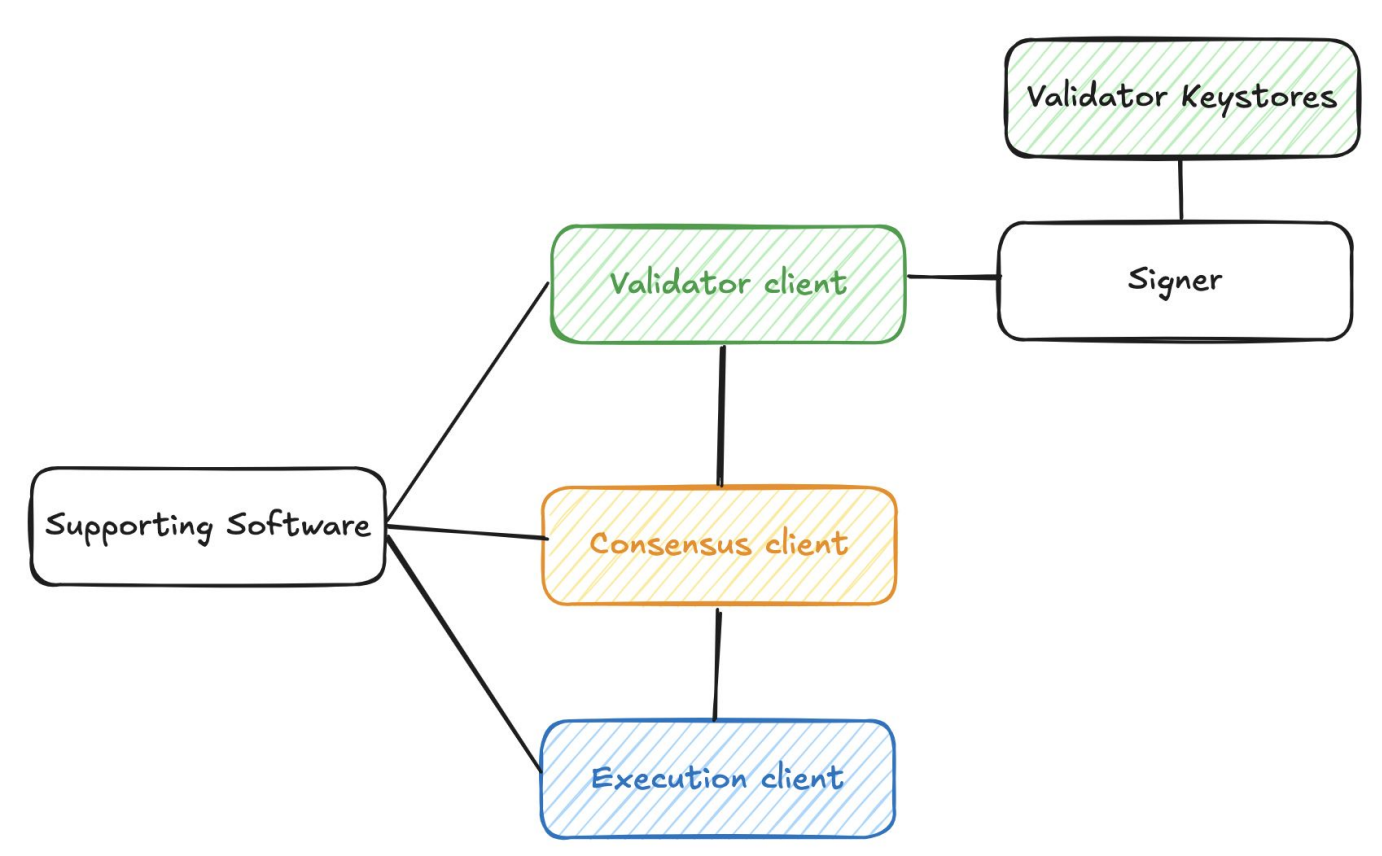
Basic Anatomy



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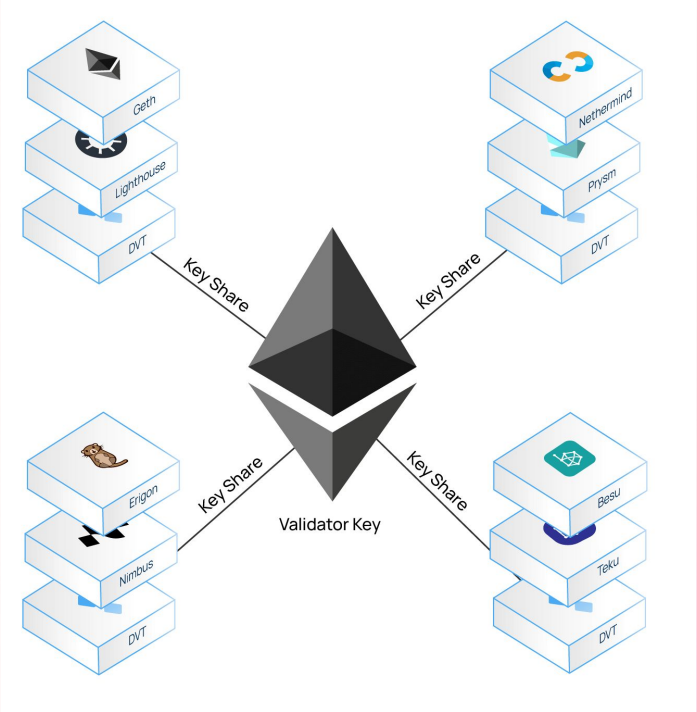
Distributed Validator Technology (DVT)

Coordination software for splitting keys and complete validator duties



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Lido CSM

1.3 ETH validators* with ~2.3x enhanced rewards

* First validator is 1.5 ETH if included in EA, 2.4 ETH if not

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Permissionless entry into Lido Node Operator set.

Our Big, Hairy, Audacious Goal is getting to 5000 independent operators!

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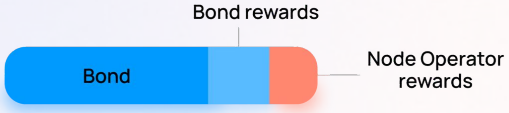
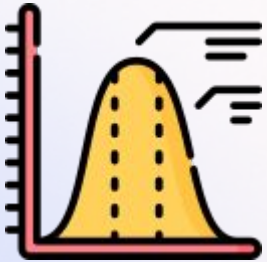
Key features

Low ETH-only bond: 1.3 ETH

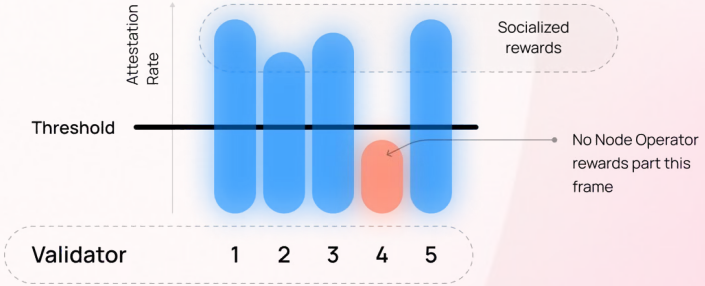


2 sources of real rewards: ~2.3x higher rewards

“Bulk discount” on bond



Mev smoothing



Rewards socialisation

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Run nodes before? You may have benefits on the Community Staking Module

https://lidofinance.github.io/csm-ea-checker/

Connect your wallet with your home staker address to see if you qualify.

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Operator requirements

- Bond serves as collateral

- Exit validators if requested by protocol

- Set withdrawal address to Lido Withdrawal Vault
`0xB9D7934878B5FB9610B3fE8A5e441e8fad7E293f` (Mainnet)

- Set fee recipient address to Lido EL Rewards Vault
`0x388C818CA8B9251b393131C08a736A67ccB19297` (Mainnet)

- Run Mev-Boost and use only Lido-vetted MEV Relays

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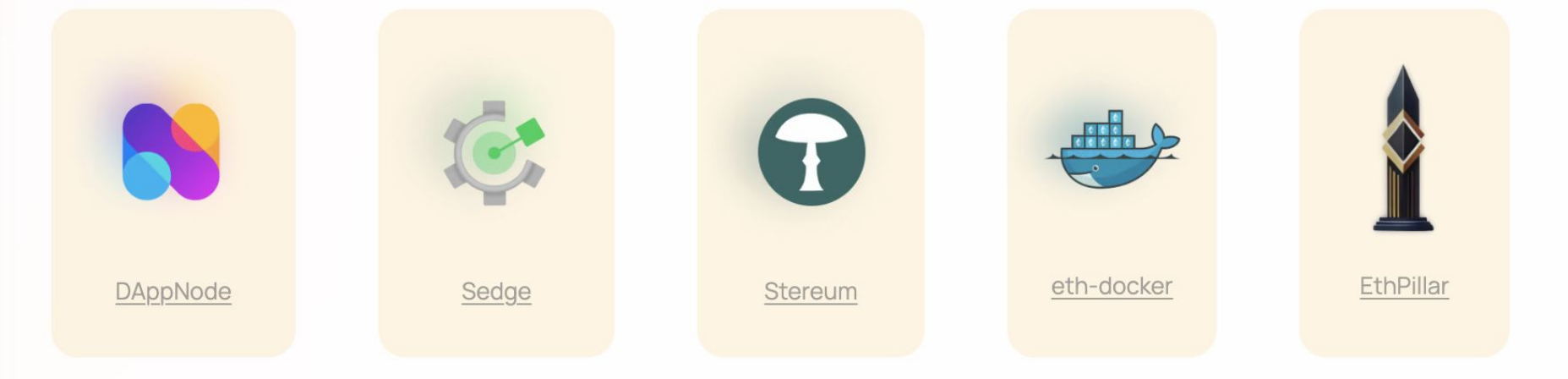
Multiple plug-and-play integrations



+ Composable with DVTs

++ Low gas fees

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Tons of community resources!

Category	Name	Maintainner/Author	Language	URL	Made by Commu
Code or Interfaces	Lido CSM Testnet Widget		English	https://csm.testnet.fi/	☐
Code or Interfaces	Lido CSM Github		English	https://github.com/lidofinance/community-staking-module	☐
Guides	Lido CSM validator setup	Sam	English	https://dvt-homestaker.stakesaurus.com/bonded-validators-setup/lido-csm	☑
Guides	CSM validator setup using ETH Pillar on Google Cloud	Sam	English	https://www.youtube.com/watch?v=aZLPACj2oPI	☑
Guides	CSM validator setup using Stereum	Sam	English	https://x.com/ma_themagician/status/1815298526835863626	☑
Guides	Validator exit	Sam	English	https://dvt-homestaker.stakesaurus.com/bonded-validators-setup/lido-csm/exiting-csm-validators#manual-exit	☑
Guides	Subscribing to CSM related events		English	https://youtu.be/ByRujrL501c	☐
Guides	CSM validator setup using EthDocker on Google Cloud	Sam	English	https://www.youtube.com/watch?v=PQ5qLfbBeTI	☑
Guides	CSM validator setup on DappNode	0xricas	English	https://www.youtube.com/watch?v=n31bSAn2luM	☑
Guides	CSM validator setup on DappNode	0xricas	Spanish	https://www.youtube.com/watch?v=oyz9yhqKldQ	☑
Guides	CSM validator setup on DappNode via X thread	Yuri	English	https://x.com/tamtamchik/status/1830230410665857300	☑

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Q&A and Discussions

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